

Hydra MAGE — Literature Verification of the Generated Shortlist

Companion to the candidate-generation supplement. **Search date:** 2026-07-16. **Model:** epoch-13563 successor.

Protocol. Each of the 22 gate-passing, plausibility-filtered showcase candidates (`shortlist_clean.csv`) was searched via web search along three tiers — exact composition as a superconductor; parent-compound + dopant; near-neighbour/family — plus a **known-magnet cross-check** for magnetic-element compositions. Verdicts: **KNOWN-SC** (this or a close variant is a reported superconductor), **RELATED** (known family, this exact stoichiometry not found), **KNOWN-MATERIAL** (exists, not reported SC), **PHASE-TRANSITION-CONFUSION** (a known magnet whose ordering the model likely mistook for Tc), **NOVEL** (no published work found). Hf-doped FeSe was included as a **calibration control**.

Honesty caveat. A web literature search is **not exhaustive** — paywalled, non-indexed, or obscure work can be missed. Every “NOVEL”/null result means *“no published work found in the searched sources (2026-07-16),”* not definitive novelty. Where the family is known but the exact doping was not clearly reported, the verdict errs to RELATED.

Verdict distribution (22 candidates)

Verdict	Count	Meaning
KNOWN-SC	10	exact / near rediscovery of a reported superconductor — validates the model’s recall
RELATED	6	known SC family, this exact doping/stoichiometry not found in searched sources
KNOWN-MATERIAL (not SC)	3	composition exists but is not a superconductor — a false positive
PHASE-TRANSITION-CONFUSION	1	a known magnet; predicted “Tc” tracks a magnetic ordering temperature
NOVEL (but chemically implausible)	2	no published work found, but the composition is a likely generator artifact
Clean, plausible, genuinely-novel survivors	0	—

Full per-candidate results

Formula	Tc pred (K)	Verdict	What the literature says	Reference
Ba _{0.2} (Fe,Ni)As ($\approx$ BaFe _{1.86} Ni _{0.14} As ₂)	8.5	KNOWN-SC	electron-doped Ba-122; composition appears verbatim	arXiv:2111.12808
Ba _{0.2} (Fe,Co)As ($\approx$ Ba(Fe _{0.905} Co _{0.095}) ₂ As ₂)	12.8	KNOWN-SC	optimal Co-doped Ba-122, Tc $\approx$ 22 K	arXiv:0807.2237

Formula	Tc pred (K)	Verdict	What the literature says	Reference
$\text{Sr}_{0.25}\text{Sc}_{0.12}\text{Fe}_{0.13}\text{P}_{0.12}\text{O}_{0.38}$ ($=\text{Sr}_4\text{Sc}_2\text{Fe}_2\text{P}_2\text{O}_6$)	16.5	KNOWN-SC	real 17 K iron oxypnictide — near-exact recall	arXiv:0903.3314
$\text{Ca}_{0.33}\text{Al}_{0.28}\text{Ga}_{0.03}\text{Si}_{0.34}$ ($\approx\text{CaAlSi}$)	6.4	KNOWN-SC	AlB_2 -type silicide SC, $T_c \approx 7.8$ K	APL 80, 1019
$\text{Mg}_{0.52}\text{Al}_{0.07}\text{Si}_{0.41}$ ($\approx\text{Mg}_4\text{Si}_3\text{Al}$)	5.0	KNOWN-SC	HP-synthesized Mg-Al-Si SC, $T_c \approx 5.2$ K	PMID 23477562
$\text{Mg}_{0.33}\text{Sc}_{0.004}\text{B}_{0.67}$ ($\approx\text{MgB}_2$)	41.9	KNOWN-SC	essentially pure MgB_2 (39 K) + trace Sc	Nature 410, 63
$\text{Mg}_{0.22}\text{Al}_{0.11}\text{B}_{0.66}$ ($\approx\text{Mg}_{0.66}\text{Al}_{0.34}\text{B}_2$)	18.5	KNOWN-SC	Al-doped MgB_2 (Al suppresses Tc)	cond-mat/0104366
$\text{Ba}_{0.22}\text{Cu}_{0.16}\text{Hg}_{0.08}\text{O}_{0.53}$ (Hg-1201)	43.1	KNOWN-SC	underdoped $\text{HgBa}_2\text{CuO}_{4+\delta}$	Nature 362, 226
$\text{Ba}_{0.08}\text{Sr}_{0.02}\text{La}_{0.13}\text{Cu}_{0.19}\text{O}_{0.56}$ (La-214)	14.5	KNOWN-SC	$(\text{La,Ba,Sr})_2\text{CuO}_4$ — the 1986 original	Bednorz–Müller lineage
$\text{Ba}_{0.14}\text{Ca}_{0.08}\text{Tl}_{0.08}\text{Cu}_{0.16}\text{Hg}_{0.02}\text{O}_{0.52}$ (Tl/Hg-1223)	124.0	KNOWN-SC	mixed Tl/Hg-1223, $T_c \approx 130$ K	MRB 29, 90566
$\text{Rb}_{0.15}(\text{Fe,Co})\text{As}$	23.7	RELATED	RbFe_2As_2 is SC; Co-doping is the canonical route; exact composition not found	arXiv:0911.2394
$\text{La}_{0.04}\text{Eu}_{0.19}\text{Fe}_{0.25}\text{As}_{0.55}$	10.3	RELATED (<i>magnetic caveat</i>)	doped EuFe_2As_2 family (SC to ~ 35 K); predicted 10 K sits below Eu AFM at 19 K	arXiv:0807.3293
$\text{K}_{0.12}\text{La}_{0.03}\text{Hf}_{0.83}$	47.9	RELATED (<i>training reproduction</i>)	LaH_x superhydride family, but $\text{K} \gg \text{La}$ is inverted and SC needs ~ 150 – 200 GPa; present in the training corpus (K–La hydride) → a reproduction, not novel	Nature 569 (LaH ₁₀)
$\text{Zr}_{0.11}\text{Ti}_{0.52}\text{Nb}_{0.24}\text{Si}_{0.11}$	11.2	RELATED	Ti-Nb-Zr HEA / Nb-Si silicide SC families; exact quaternary not found	PNAS 113 (2016)
$\text{La}_{0.74}\text{Si}_{0.13}\text{Ni}_{0.07}$	11.4	RELATED	LaNiSi is SC but only ~ 1.2 K; predicted 11 K implausibly high for this La-rich phase	PRB 50, 6523
$\text{Sr}_{0.06}\text{Ca}_{0.15}\text{Cu}_{0.25}\text{Bi}_{0.13}\text{O}_{0.45}$ (BSCCO)	12.3	RELATED	BSCCO family; Sr-deficient off-standard stoichiometry	—
$\text{Nb}_{0.28}\text{In}_{0.01}\text{Cu}_{0.70}$	9.1	KNOWN-MATERIAL (<i>training reproduction</i>)	Nb–Cu are immiscible; T_c $9.1 \approx$ elemental Nb (9.26 K) — an elemental-Nb echo , not a compound; present near-verbatim in the training corpus → a reproduction, not novel	—
$\text{Al}_{0.19}\text{B}_{0.52}\text{C}_{0.19}$	9.2	KNOWN-MATERIAL	Al-B-C ceramics (Al_3BC etc.) exist; none reported SC; AlB_2 lacks MgB_2 's hole-doped σ -bands	—
$\text{Ca}_{0.04}\text{Y}_{0.21}\text{Tm}_{0.07}\text{O}_{0.58}$	13.4	KNOWN-MATERIAL	an <i>insulating</i> $(\text{Y,Tm})_2\text{O}_3$ -type sesquioxide — no SC and no ordering near 13 K (spurious head output)	—
$\text{Eu}_{0.28}\text{Fe}_{0.22}\text{S}_{0.49}$	10.5	PHASE-TRANSITION-CONFUSION	Eu-rich sulfide; Eu^{2+} magnetic ordering (cf. EuS ferro at 16.6 K) falls in the predicted- T_c band	Nature Mater. (BaFe ₂ S ₃)
$\text{Fe}_{0.31}\text{Ni}_{0.05}\text{As}_{0.36}\text{Hf}_{0.25}\text{Se}_{0.18}$	17.1	NOVEL (artifact)	no mixed-anion Fe-As-Se-H phase exists; H+Ni-on-Fe = likely generator artifact	—
$\text{Mg}_{0.52}\text{Sn}_{0.25}\text{B}_{0.23}$	28.6	NOVEL (artifact)	Sn does not substitute into the MgB_2 σ -lattice; B-poor/Sn-rich = structurally implausible	—

Key findings

1. **No clean, plausible, genuinely-novel survivor.** The literature search of the whole 22-candidate shortlist promotes **zero** additional predictions to Hf-FeSe’s standard.
2. **The model is a strong recaller of known superconductivity.** Ten candidates are exact/near rediscoveries — including a **near-exact match to the real 17 K oxypnictide $\text{Sr}_4\text{Sc}_2\text{Fe}_2\text{P}_2\text{O}_6$** — which is a genuine validation of the model’s compositional memory across cuprate, iron-based, MgB_2 -type, and silicide families.
3. **“Database-novel” $\neq$ “literature-novel” $\neq$ “plausible.”** A direct canonical-composition check against the training corpus partitions the novelty-flagged shortlist cleanly: **K-La-H and Nb-In-Cu (the latter near-verbatim) are in fact present in the training set** — reproductions, not novel-ties — while **six compositions are genuinely absent from the training corpus** ($\text{Rb}(\text{Fe},\text{Co})\text{As}$, Zr-Ti-Nb-Si , Al-B-C , the (Y,Tm)-oxide, Eu-Fe-S , Fe-Ni-As-Se-H), yet **none is a confirmed superconductor**: on literature search they resolve into RELATED known-family variants ($\text{Rb}(\text{Fe},\text{Co})\text{As}$, Zr-Ti-Nb-Si), known **non-superconductors** (Al-B-C , the Tm-oxide), a **magnetic confusion** (Eu-Fe-S), and a **chemically-implausible artifact** (Fe-Ni-As-Se-H).
4. **The flagship carries a plausibility caveat too.** Hf-FeSe reproduces its “no published work found” verdict, but Fe-site transition-metal substitution is documented to *suppress* FeSe superconductivity — so even the flagship is a hypothesis with a real physical objection, not a confident prediction.

Bottom line: the generation loop’s value is a **disciplined recall/interpolation of known SC chemistry**, not a source of many new materials. The gate + novelty + literature pipeline is what separates the one flagged hypothesis from the rediscoveries and the artifacts — which is the paper’s calibration/honesty thesis, confirmed at the sharpest test.